

Engineering Mathematics 1

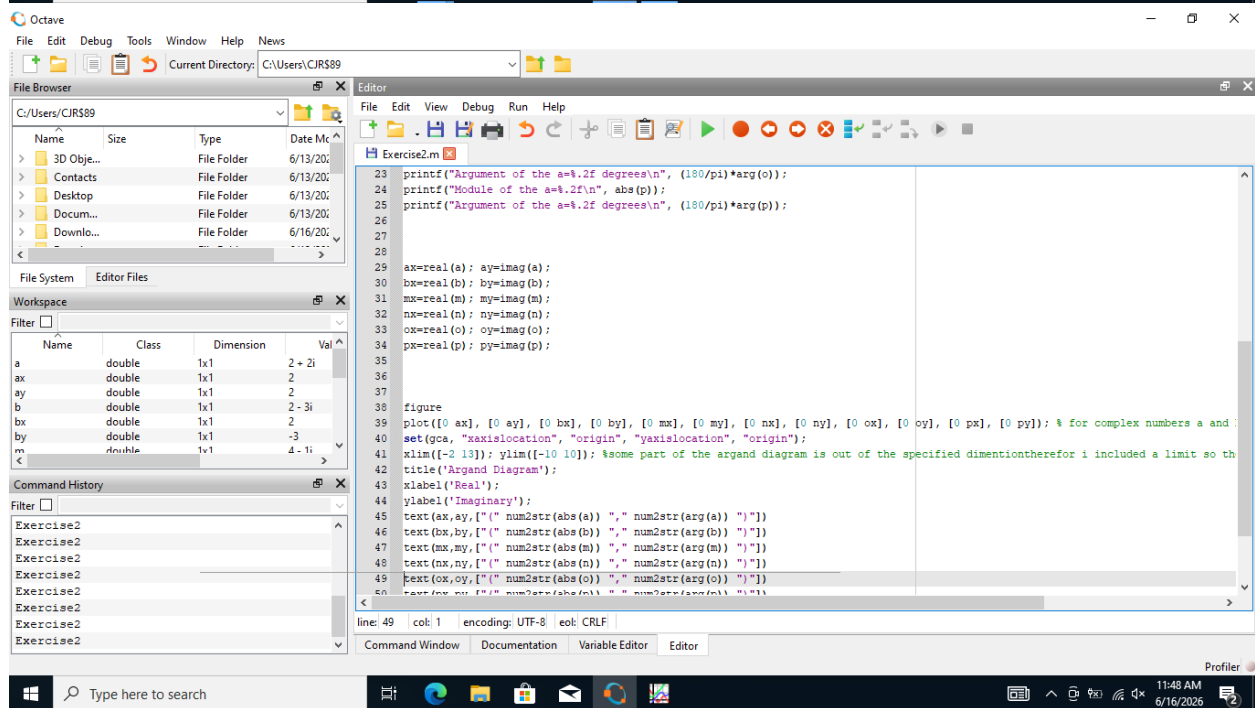


STD Number – IT26101833

STD Name – Rajapaksha R.A.C.J

Lab 02;

Exercise 2:



Octave

File Edit Debug Tools Window Help News

Current Directory: C:\Users\CJRS89

File Browser

C:\Users\CJRS89

Name	Size	Type	Date Modified
3D Objects		File Folder	6/13/2020
Contacts		File Folder	6/13/2020
Desktop		File Folder	6/13/2020
Documents		File Folder	6/13/2020
Downloads		File Folder	6/16/2020

File System Editor Files

Workspace

Name	Class	Dimension	Value
a	double	1x1	2 + 2i
ax	double	1x1	2
ay	double	1x1	2
b	double	1x1	2 - 3i
bx	double	1x1	2
by	double	1x1	-3
m	double	1x1	4 - 1i

Command History

Filter

Exercise2
Exercise2
Exercise2
Exercise2
Exercise2
Exercise2
Exercise2
Exercise2

Editor

```

29 ax=real(a); ay=imag(a);
30 bx=real(b); by=imag(b);
31 mx=real(m); my=imag(m);
32 nx=real(n); ny=imag(n);
33 ox=real(o); oy=imag(o);
34 px=real(p); py=imag(p);
35
36
37
38 figure
39 plot([0 ax], [0 ay], [0 bx], [0 by], [0 mx], [0 nx], [0 ny], [0 ox], [0 oy], [0 px], [0 py]); % for complex numbers a and b
40 set(gca, "xaxislocation", "origin", "yaxislocation", "origin");
41 xlim([-2 13]); ylim([-10 10]); %some part of the argand diagram is out of the specified dimensiontherefore i included a limit so th
42 title('Argand Diagram');
43 xlabel('Real');
44 ylabel('Imaginary');
45 text(ax,ay,[" num2str(abs(a)) ", " num2str(arg(a)) "]);
46 text(bx,by,[" num2str(abs(b)) ", " num2str(arg(b)) "]);
47 text(mx,my,[" num2str(abs(m)) ", " num2str(arg(m)) "]);
48 text(nx,ny,[" num2str(abs(n)) ", " num2str(arg(n)) "]);
49 text(ox,oy,[" num2str(abs(o)) ", " num2str(arg(o)) "]);
50 text(px,py,[" num2str(abs(p)) ", " num2str(arg(p)) "]);
51 grid('on');
52 legend('a', 'b', 'm', 'n', 'o', 'p');
53
54
55

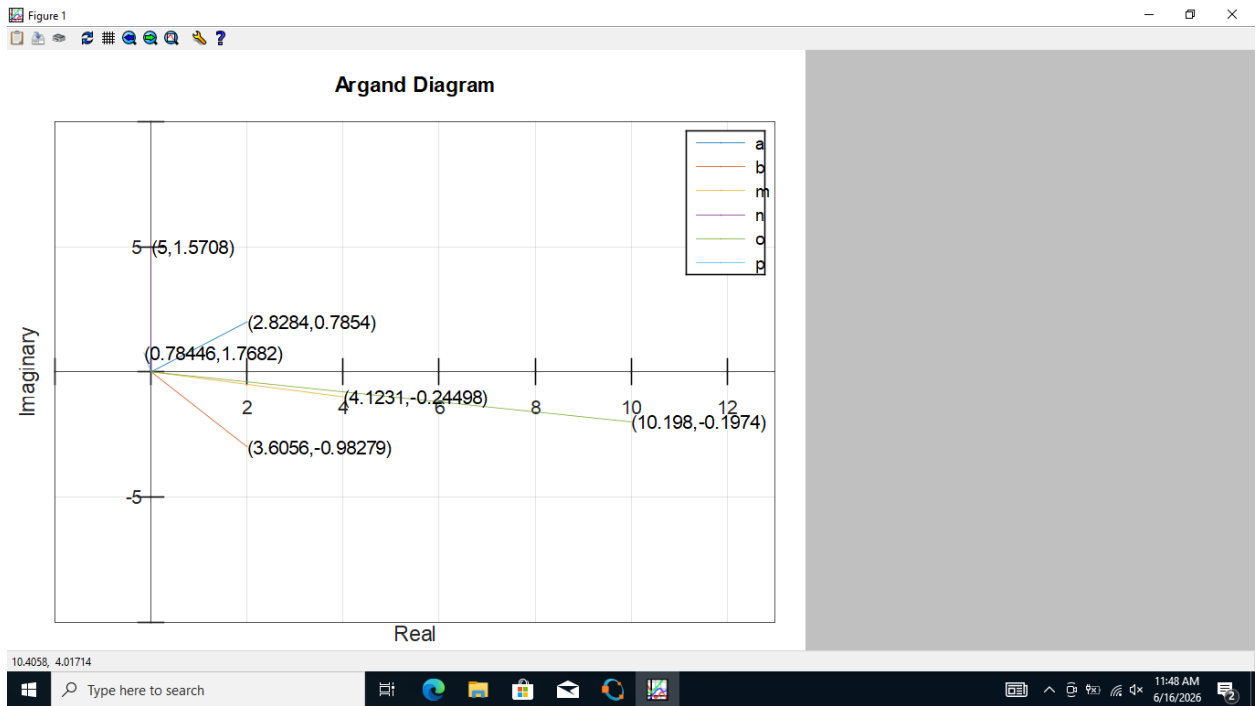
```

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Command Window Documentation Variable Editor Editor

Profiler

11:48 AM 6/16/2020



Octave

File Edit Debug Tools Window Help News

Current Directory: C:\Users\CJR589

File Browser

C:\Users\CJR589

Name	Size	Type	Date Modified
>			
3D Objects		File Folder	6/13/2020
Contacts		File Folder	6/13/2020
Desktop		File Folder	6/13/2020
Documents		File Folder	6/13/2020
Downloads		File Folder	6/16/2020

File System Editor Files

Workspace

Name	Class	Dimension	Value
a	double	1x1	2 + 2i
ax	double	1x1	2
ay	double	1x1	2
b	double	1x1	2 - 3i
bx	double	1x1	2
by	double	1x1	-3
m	double	1x1	4 - 1i

Command History

Filter

Exercise2
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Exercise2
Exercise2
Exercise2
Exercise2
Exercise2
Exercise2

Command Window

```
Module of the a=2.83  
Argument of the a=45.00 degrees  
Module of the b=3.61  
Argument of the b=-56.31 degrees  
Module of the a=4.12  
Argument of the a=-14.04 degrees  
Module of the a=5.00  
Argument of the a=90.00 degrees  
Module of the a=10.20  
Argument of the a=-11.31 degrees  
Module of the a=0.78  
Argument of the a=101.31 degrees  
octave:21> |
```

Command Window Documentation Variable Editor Editor

Profiler

11:48 AM
6/16/2020

